

Assessment

Module 6 Java - Core Java

Module 7 - Java - RDBMS & Database Programming With JDBC

Case Overview

Scenario :

You are a software engineer at **SmileCare Dental Services**, a busy urban dental clinic. Their current appointment and billing system is paper based, leading to scheduling conflicts, lost treatment records, and manual billing errors. The clinic has requested a Java-based solution to digitize and simplify their operations:

Core functionality	
1. Appointment Booking	- Allows reception staff to book appointments for patients. - Captures type of dental service (e.g., cleaning, filling, root canal), preferred date/time, and estimated cost.
2. Billing	- Automatically generates invoice after treatment. - Calculates total charges based on treatment type, medications, taxes, and insurance discount. - Exports the invoice to a GUI component and stores it in the patient's service history.

Key Competencies Tested:

Java Concept: OOP, Exception, File/Database, Swing, Report Generator

Practical considerations: data persistence with RDBMS/Excel, input validation, and realistic healthcare workflows.

Reflective thinking: Ability to scale the system for multi-branch clinics or patient follow-up alerts.